Customer Shopping Behavior Analysis

3,900 purchases | 18 columns | goal: uncover spending patterns, segments, preferences, subscriptions

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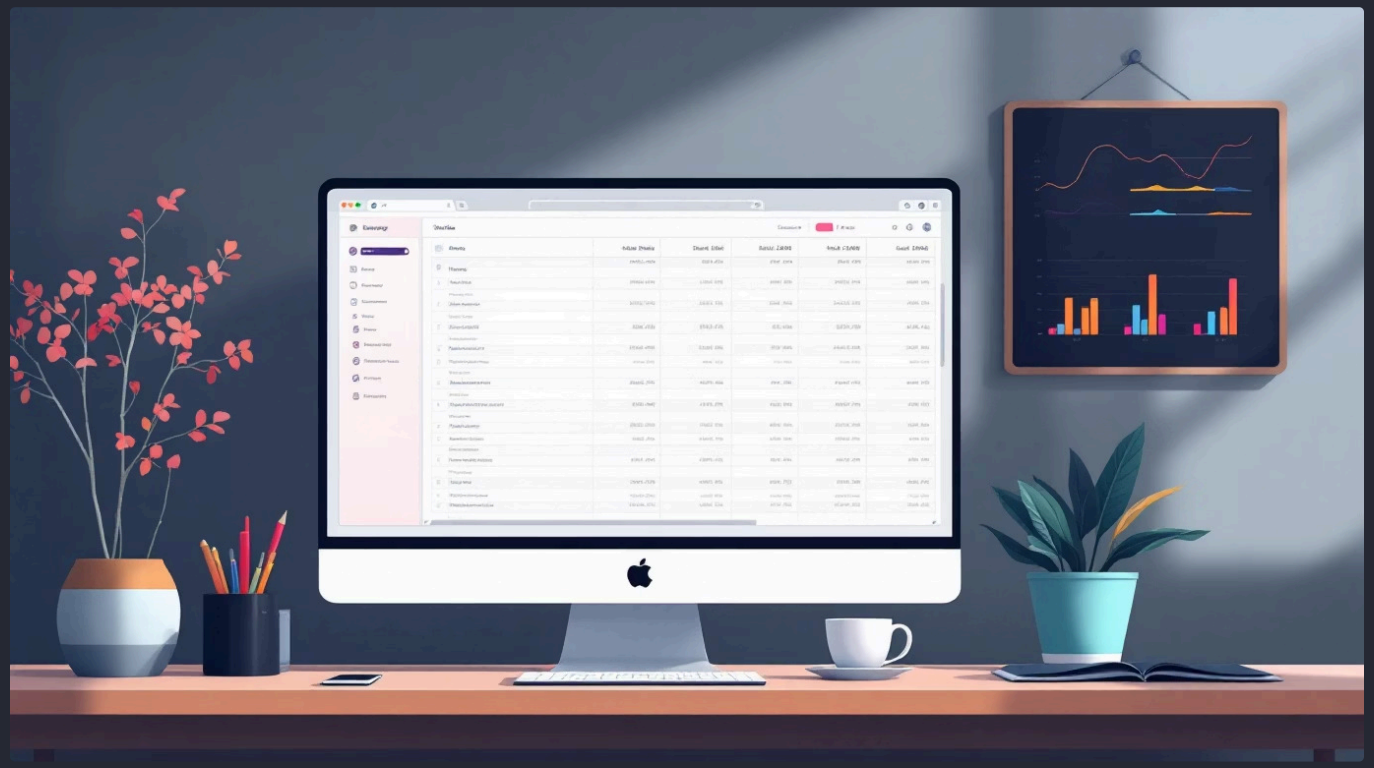
Dataset Snapshot



Rows

3,900

Columns

18

Missing

37 values in Review Rating

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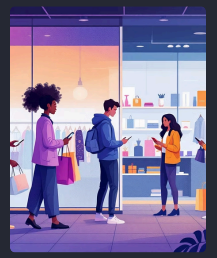
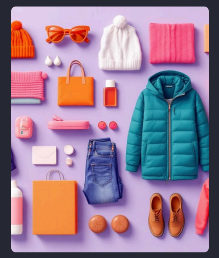
Key Features



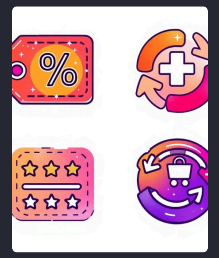
Demographics

Age, Gender, Location, Subscription Status



Purchase Details

Item, Category, Amount, Season, Size, Color



Behavior

Discounts, Promo Codes, Frequency, Ratings, Shipping

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Python EDA Highlights

01

Load & Inspect

02

Missing Values

03

Feature Engineering

04

Clean-up

05

DB Load

Cleaned DF → PostgreSQL

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SQL Business Analysis — Key Results

Revenue by Gender

Female: 75,191 | Male: 157,890

High-Spend Discount Users

839 rows identified (examples: customer 4 → $90, 24 → $99)

Top Products by Rating

Gloves 3.86, Sandals 3.84, Boots 3.82, Hat 3.80, Skirt 3.78

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Shipping & Subscription Insights

Shipping Avg Spend

Standard: $58.46 | Express: $60.48

Subscribers

Yes: 1,053 customers, avg $59.49, revenue $62,645

Non-Subscribers

No: 2,847 customers, avg $59.87, revenue $170,436

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Discount-Dependent Products

Products with highest discounted purchase rates:

- Hat — 50.00%
- Sneakers — 49.66%
- Coat — 49.07%
- Sweater — 48.17%
- Pants — 47.37%

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Customer Segments & Top Items

Top items by category (examples):

- Accessories: Jewelry, Sunglasses, Belt
- Clothing: Blouse, Pants, Shirt
- Footwear: Sandals, Shoes, Sneakers
- Outerwear: Jacket, Coat

Loyal

3,116 customers

Returning

701 customers

New

83 customers

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Dashboard Snapshot

KPIs: 3.9K customers | $59.76 avg purchase | 3.75 avg rating

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Business Recommendations

Boost Subscriptions

Promote exclusive subscriber benefits

Loyalty Programs

Reward repeat buyers to increase loyalty

Review Discount Policy

Balance sales lift with margin control

Product Positioning

Highlight top-rated & best-selling items

Targeted Marketing

Focus on high-revenue age groups & express-shipping users

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